



PARTICLE PHYSICS

What Is Dark Matter?

**More experts than ever think
we might never know
what it really is**

BY JOSEPH HOWLETT



Photomultiplier tubes in the XENONnT experiment at Italy's Gran Sasso National Laboratory search for signs of dark matter particles interacting with liquid xenon.

USED TO HAVE A MORE INTERESTING MORNING COMMUTE. At daybreak I'd squeeze into a tiny Fiat with all the physicists it could fit. We'd drive through winding Italian roads and onto a highway that cuts across the peninsula, with a tunnel right through the country's grandest mountain, Gran Sasso. At the tunnel's deepest point, with nearly a mile of bedrock resting above our heads, we'd turn off through a guarded gate and enter one of the quietest places on Earth.

We were there, in 2019, to build a particle detector more sensitive than any in history. We spent long workdays, overnight shifts and most weekends in this cavern, locked in a race with international rivals to find the missing five sixths of the universe's matter.

In spare moments while turning a wrench or staring at a pressure gauge, I'd sometimes ask my teammates what our chances were of winning that race. "Zero," they'd usually reply. Surprised, I began questioning colleagues beyond my several hundred collaborators, polling them at conferences and happy hours, and found that it wasn't just us. Few experts—among the younger generation basically no one—thought any scientific effort was on the brink of discovering dark matter.

In the seven years since, these murmurs have grown only louder. For four decades now we've failed to identify the particles that make up dark matter—the invisible chunk of the universe that we know is there, that suffuses every galaxy we see. And more physicists than ever are conceding that this situation might never change. "Nature doesn't always promise us the results we'd like," says physicist and mathematician Edward Witten, whose 1985 paper proposing that weakly interacting massive particles, or WIMPs, could be observed on Earth launched the field that I focused my Ph.D. work on. "It's unfortunate, but it's certainly possible that nature chose a dark matter particle that we won't be able to detect," Witten observes.

Many believers remain. Dark matter hunters are loping desperately in a host of directions both old and new, hoping that nature can't keep this secret forever. But I've begun to wonder where their optimism comes from and to question one of my deepest convictions: that the scientific method, given enough time and smart people, will get us over any hurdle, bringing us closer to the truth of things.

If dark matter disproves this conviction—if five sixths of all matter will continue to hide for the remainder of human history behind an impenetrable wall of unknowing—then maybe I've been wrong about science itself all along. Maybe all scientists should be grappling with a possibility they've long ignored but which many philosophers have long since accepted: That science will never give us a complete picture of reality. Or, to take this idea to its extreme—as I've begun to do in my darker moments—that our theories are merely a useful but ultimately meaningless facsimile of the world. Dark matter, I've come to believe, could thus become science's greatest lesson.

THE FIRST PROBLEM is that dark matter definitely exists. Astronomer Vera C. Rubin spent the 1970s meticulously cataloging the swirl of as many galaxies in the sky as she could. What she found was that the stars near the edge of every galaxy in the observable universe are moving way too fast. Like a racecar hitting a corner with too much speed,

these stars should be sent flying off their orbital racetracks and into the void. The gravity of everything we can see at the center of their galaxies simply isn't enough to hold them in.

Rubin's data were airtight. They left two alternatives: either gravity doesn't work the way physicists thought it did, or there's a heap of matter in every galaxy that we can't see.

The intervening decades have all but killed the former possibility. Astronomers construct gravitational "maps" of galaxy clusters based on tiny deflections of the light passing through those galaxies caused by the matter they contain. Most of this matter exists as a giant, invisible cloud superimposed on the stuff we can see with telescopes.

These dark matter maps confirm Rubin's observation: the gravity of these galaxies is about five times stronger than their visible stars and dust would suggest. And when astronomers caught two of these clumps of galaxies colliding, they found a powerful confirmation that it was matter, not gravity, that we had been wrong about.

The cataclysm, called the Bullet Cluster, stripped each clump's regular matter from its dark, superimposed cloud. The gas, stars and galaxies slowed down, roiled into a hot maelstrom. But the two ghostly halos of invisible stuff sailed right past each other, leaving their visible counterparts behind. If the solution to Rubin's mystery is a new law of gravity rather than extra stuff, then what's in those fleeing clouds?

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Two galaxy clusters collided in the Bullet Cluster, where the regular matter (pink) has been stripped away from the dark matter (blue).

The Bullet Cluster is nigh impossible to explain by modifying Isaac Newton and Albert Einstein's theories of gravity. So are other recent observations, such as galaxies with anomalously slight or severe "racecar" problems—there can be more or less missing matter in different places, but gravity has to look the same everywhere.

Even in our oldest snapshot of the universe's early contents, the so-called cosmic microwave background, dark matter's influence is unmissable. A few hundred thousand years after the big bang, "regular matter"—the 16 species of particles that make up the Standard Model of particle physics—was still a swirling soup, too hot to clump under gravity's pull. But the data reveal that dark matter was already starting to coalesce.

All this evidence led physicists to a simple, coherent, frustrating explanation. Five sixths of the universe's matter is a kind we can't see, yet it built everything we can see, and it's passing through us every moment. But what the heck is it made of?

Fortunately, theorists had an answer ready. Their favorite theory of matter—called supersymmetry—already predicted an as yet undiscovered particle, some 50 times the mass of a proton, that

could effortlessly explain every apparent contradiction logged by Rubin and her successors. WIMPs so perfectly addressed this cosmic gap that they became known as the "WIMP miracle." And even more miraculously, we might be able to detect them.

WIMPs would interact with regular matter mainly through their collective gravitational pull. But the individual particles also had a minuscule bridge to the Standard Model: they could interact with its particles through the weak force. This meant a WIMP passing through a block of regular matter would usually whiz through like a ghost, but very rarely this ghost would become momentarily corporeal—the WIMP would collide with an atom.

In 1985 Witten and Mark Goodman proposed a simple-sounding idea: fill a tank with atoms that flash when struck, put it underground where cosmic rays can't light it up, and wait for dark matter particles to make their identity known. It seemed like the logical next step, the way things had worked throughout the 20th century, when physicists used similar detectors to pin down each of the Standard Model's characters one by one. Dark matter, physicists figured, would pan out no

differently than earlier mysteries, such as how spin works or what protons and neutrons are made of. We'd trap the mysterious entity in a laboratory and then carefully tease out its workings.

"I was giving talks, saying, 'We're going find this in 10 years,' and everybody thought that was true," says Katherine Freese, a theoretical astrophysicist at the University of Texas. Freese co-authored a 1986 paper that filled in the details of Goodman and Witten's proposal, establishing "direct" dark matter detection as a field. "We were really convinced, like, 'My God, we can do this,'" she says. "We thought it was right around the corner then."

I JOINED THE ARMS RACE near its peak, almost two decades after Freese and her cadre sparked it. New experiments were launching every year, battling to cover the full span of what the WIMP's mass and cross section (precisely how ghostly it is) could possibly be.

One technology had taken a clear lead: detectors filled with liquid xenon, pioneered by Columbia University physicist Elena Aprile. It's hard to point to a single person more responsible for our tearing through so much of the WIMP's massive parameter space so

quickly. “She really is the one who got the use of xenon going,” Freese says. So I asked Aprile to be my doctoral adviser in 2015. To me, dark matter was the biggest embarrassment in physics, and I couldn’t imagine a higher priority than using the scientific method to fix it.

The first blow to this resolve came in 2018. We had in hand an unprecedented chunk of data from the most sensitive experiment in history, XENON1T. It was poised to probe a great swath of the remaining masses and cross sections the WIMP could still have. Using fake data to blind ourselves from potential bias, we’d built up an elaborate system to automatically analyze the real thing. Now our entire collaboration had joined a conference call to press a button and watch the results unfold together. We saw our years of work collapse in seconds. I’ll never forget the sight of despair washing across a Zoom screen full of faces as our search turned up nothing.

When I went to Italy to help build that experiment’s successor, XENONnT, I started polling my peers and soon realized how little of the confidence Freese spoke of remained. That work, too, would go on to reveal nothing, as did its many competitors.

Physicists are already planning another round of experiments, based on theorists’ updated estimates of just how weakly interacting WIMPs can be. But the original miracle they pulled from their desk drawer is dead. These modified WIMPs must be tethered even more loosely to the Standard Model, by way of some additional, unknown particles.

But even if this updated guess is right, there’s a good chance we’ll never find what we’re looking for. Our bodies and underground detectors alike are constantly pummeled by billions of neutrinos produced in the sun and atmosphere. Neutrinos interact far more rarely than the original WIMP but similarly to or more often than any version of WIMPs that survived the 2010s. And their interactions are impossible to distinguish

from dark matter’s. Xenon experiments have already begun to see traces of this impenetrable “neutrino floor.”

So the community has moved on to other hypotheticals—to axions, a long-standing alternative hypothesis theorists have had ready-made since 1977, or to lower-mass WIMPs whose tiny bump wouldn’t light up a xenon atom. People are paying new attention to the possibility that dark matter is made of primordial black holes, hypothetical entities dating to the big bang that Stephen Hawking predicted in 1971. Each of these black holes swirling around the galaxy would have a mass maybe a bit less than the moon’s packed into a microscopic ball.

But what if dark matter is none of these things? What if it is not connected to our world through any force besides gravity? In that case, no conceivable technology could sense a single particle

of the stuff. Dark matter would be like a shadow realm, superimposed on but disconnected from our own except through its collective tug, its denizens utterly unknowable to us.

CALL ME NAIVE, but I never thought this could happen. The scientific method is built to encounter, interrogate and ultimately subsume any mystery it runs across. So I figured dark matter was just another speed bump. Give us enough time, and we’d figure it out like we had so many previous puzzles.

In retrospect, all those recent wins might have misled me. “Particle physics during the 20th century was incredibly successful,” says Simon Allzén, a philosopher at Stockholm University. At blackboards and colliders, my predecessors had conquered and cataloged a zoo of particles into the elegant, orderly



In 2019 the author and other physicists close the chamber that will house the XENONnT experiment at the Gran Sasso lab.

Elena Aprile

Standard Model. They came to expect a “five-sigma” standard of evidence, meaning a true discovery can have only a one-in-several-million chance of being a statistical fluke. Maybe that bar is just too high to maintain. “This has been a golden era,” Allzén says. “Modeling our scientific method on what we achieved with the Standard Model is a bad idea.”

Stop exalting physics above the other sciences—after all, as Allzén puts it, we believe in dinosaurs even though we can’t trap one in a lab—and maybe dark matter isn’t such a problem. “We have detected it,” argues Siska De Baerdemaeker, a philosopher at Stockholm University. “That you haven’t found something until you’ve found it in a lab is weirdly dismissive of gravitational detection.”

Dark matter is there. Using just its gravity, we’ve built extensive catalogs of the shadowy stuff’s presence and evolution across the eons. Maybe we should accept that the secondary goal—figuring out what its constituents are—might be beyond reach. “That would be the worst-case scenario for physics, especially for particle physics,” De Baerdemaeker says. “But nature doesn’t have to play nice.”

Still, I can’t quite shake my training. Atoms are made up of electrons and nucleons, which consist of up and down quarks. How can we call science a success if we’ll never know what this other stuff—which is almost all of the stuff—is made of?

Dark matter has me thinking I’ve been wrong about what science is even for. I thought it was the ultimate place for asking humankind’s biggest questions (How did we get here? What is everything made of? How does it all come together?) because it was the only place where you could get answers that were simply true. But are they?

Every theory we’ve ever had has technically been wrong—an imperfect approximation of reality—and we can be sure the Standard Model is no different. Dark matter is just the most glaring of its gaps, different only in degree from its other fuzzy impressions of the world, from the electron to the Higgs boson.

I’m starting to think none of these objects are any more “real” than the



The author (center) and other scientists huddle around a computer screen far below the Gran Sasso mountain, watching the results of a critical equipment test.

WIMP. They’re just theoretical ideas that help us predict what will happen, like how high school-level mechanics can predict how far away a punted football will land. I’m starting to think that is science’s actual job: predicting what will happen. The Standard Model is just a fancier version of those high school equations, a reliable framework for saying what will come out when you smash matter together at high speeds. Maybe the search for absolute truth—not just a reliably predictive picture of the world but a meaningful one—was always too ambitious.

Tim Maudlin, a philosopher of science at New York University, says that I’ve retreated too far, that I should “lighten up a little.” Our theory may never be completely finished, he points out, but that doesn’t mean entities such as the electron have zero connection to reality. But Maudlin also warns against my former unassailable optimism, calling it a kind of “theological thesis.” There may be fundamental or practical limitations that stop us from ever answering a scientific question. “I mean, that’s life, buddy,” he says. “The world doesn’t care about you.”

Something like faith might be what it actually takes to be a scientist, even if it isn’t totally rational. In fact, most of the

physicists I’ve talked to still believe we’ll make sense of dark matter someday. “I do think we’ll get it,” Freese says. “Just because we haven’t solved a problem doesn’t mean the next generation won’t be smart enough to figure it out.” Aprile, my former adviser, thinks the right stroke of experimental genius is all that’s lacking. “We have to invent a tool. The progress has always only been made because Galileo invented the telescope,” she says. “Some smart kid one day will figure it out. I know it must be.”

Scientists have never really been motivated by a desire to predict where the football will land. They’re dreamers in search of answers, of meaning, whether it’s there or not, even if they have to strive toward a receding horizon. “What else is there—give up? And then what? Do something else?” Aprile asks me these questions rhetorically. Her path was never a choice. “I’m never happy, actually—that’s the problem,” she says. “Ever since I was young, I wanted to get out, I wanted to do more.”

Someone once told me that the most beautiful thing about science is how it points at its own limitations. Maybe that’s the beauty of dark matter. Maybe Aprile and the other restless dreamers have dug up the ultimate limit—one the rest of us can’t ignore. ●